

Computer Networking

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Introduction

- **Packets:** A packet is a formatted unit of data carried across a packet-switched network

Divided into a **header** (control info like source/destination addresses and sequence numbers) and a **payload** (the actual user data).

Large files or messages are broken down into these smaller chunks so they can share network paths efficiently before being reassembled at the destination

- **Packet switch:** A packet switch is a hardware device that takes a packet arriving on an incoming communication link and forwards it toward its destination on an outgoing link

Routers (which manage traffic across different networks) and link-layer switches (which manage traffic within a local network).

- **Communication links:** A communication link is the physical or wireless medium that connects devices and packet switches together
 - **Transmission Rate:** Measured in bandwidth or throughput, determining how fast data can be pushed through the link.
 - **Types:** Can be point-to-point (dedicated connection between two nodes, like an Ethernet cable) or broadcast (shared medium where multiple nodes can receive the transmission)

A communication link in networking is a physical or logical channel that connects two or more devices to transmit data

- **Distributed applications:** A distributed application is software with components running on multiple independent computers across a network that communicate and coordinate to function as a single system
 - **Nodes:** Separate computers or servers running parts of the application.
 - **Network Messaging:** The way nodes talk to each other using data formats like JSON or XML.
 - **Middleware:** A software layer that handles routing, security, and communication between different machines
- **Socket interface:** A network socket interface is an endpoint that allows two programs to pass data back and forth across a computer network
 - A network socket acts as a virtual door for network communication.
 - It combines an IP address (to find the computer) and a port number (to find the specific app).

An IP address is like a street address, and a port is like an apartment or room number within that building.

- **IP Address (The Building):** This gets the data to the correct physical location or device on the network, just like a street address gets a letter to a specific apartment building.
- **Port (The Apartment Number):** Once inside the building, the port tells the data exactly which door to go to (such as a specific app, service, or program running on that device).

- **Protocol:** A networking protocol is a set of established rules that allows different electronic devices on a network to communicate with each other.
 - How data is packaged into smaller pieces called packets.
 - How data is addressed so it reaches the correct destination.
 - How errors are detected and fixed if data goes missing.

Think of a networking protocol like a standard phone call conversation:

1. **The Setup:** You dial a phone number (the network address) to signal you want to connect.
2. **The Greeting:** The other person answers and says "Hello" (establishing a connection).
3. **The Exchange:** You speak in a shared language like English, taking turns so you do not talk over each other (data transmission rules).
4. **The Goodbye:** You both say goodbye and hang up to end the call (terminating the connection)

If one person spoke only code and the other spoke Spanish, or if both screamed at the exact same time without listening, the conversation would fail. A networking protocol prevents this failure for computers.

- **End systems / hosts:** In computer networking jargon, the computers and other devices connected to the Internet are often referred to as end systems. They are referred to as end systems because they sit at the edge of the Internet

End systems are also referred to as hosts because they host (that is, run) application programs such as a Web browser program, a Web server program, an e-mail

- **IoT:** The Internet of Things (IoT) is a giant network of physical objects embedded with sensors, software, and other technologies that connect and share data with other devices and systems over the internet

The phrase "Internet of Things" was coined by Kevin Ashton in 1999. He used the phrase as the title of a presentation he gave to pitch a new supply-chain optimization project that used radio-frequency identification (RFID) tags to track lipsticks and other products. He wanted to bridge the gap between the physical world ("things") and the digital world ("the Internet")

Note: Companies routinely embed and conceal RFID tags inside product packaging, care labels, and fabric seams for inventory tracking and loss prevention

Radio Frequency Identification, which is a wireless technology that uses radio waves to identify, track, and communicate information about objects, animals, or people. The range of an RFID (Radio Frequency Identification) system spans from a few centimeters to over 100 meters

- **Cloud computing:** Cloud computing in networking is the delivery of network capabilities, computing power, and data storage as on-demand services over the internet instead of using physical, local hardware

Software abstracts physical hardware into virtual pools of resources like Virtual Private Clouds (VPCs) and virtual routers.

- **Compute:** In technology and cloud computing, compute refers to the processing power, memory, and hardware resources needed to run software, execute instructions, and process data
- **Clients and servers:** In a computer network, a Client-server model divides tasks between service requesters (clients) and service providers (servers)
- **Data center:** A data center is a physical facility or building that houses computer systems, servers, and related hardware to store, process, and share digital information and applications
- **Hertz:** A hertz (abbreviated as Hz) is the standard unit of measurement for frequency, equal to one cycle or wave per second

One hertz means an event repeats one time every single second

- **Data stream:** In a data stream, Hertz (Hz) matters because it measures the sampling rate or update frequency, which directly determines how much data is captured, how real-time the stream is, and the resources required to process it.

One Hertz equals one data point or update per second.

- **Telco (Telecommunications company):** Telcos own and manage the large-scale physical infrastructure like utility poles, underground conduits, and switching centers—that connect homes and businesses to the broader network.
- **Digital subscriber line:** An internet technology that sends digital data over traditional copper telephone lines.
- **Fiber-optic internet:** A high-speed connection that transmits data as pulses of light through thin strands of pure glass or plastic

Lasers or LEDs send light signals down the glass fiber, which are converted into electrical data by an Optical Network Terminal (ONT) at your home

- **Modem (modulator-demodulator):** A modem is a hardware device that connects your home or office network to the internet through your Internet Service Provider (ISP)
 - **Modem:** Connects your house to the outside internet. It usually has only one port for an Ethernet cable.
 - **Router:** Takes the internet connection from the modem and shares it with multiple devices in your home via Wi-Fi or extra Ethernet cables.
 - **Gateway:** A combo device that acts as both a modem and a router in one box

The name modulator-demodulator describes the two primary jobs the device performs to allow computers to talk to each other over long distances.

- **Modulator (Mo):** Computers speak in digital data (binary code made of 1s and 0s). However, traditional transmission lines—like old telephone wires—were designed to carry analog data (sound waves). The modem modulates (converts) the digital data from a computer into an analog signal that can travel across the network.

- **Demodulator (Dem):** When those analog signals reach the receiving end, the computer on that side cannot understand them. The modem demodulates (converts) the incoming analog waves back into digital 1s and 0s that the computer can read.

Once the ISP demodulates the analog signal back into digital data, it processes that data to get it to its final destination on the global internet.

The ISP's routers read the "destination address" attached to your data packets. The router then determines the fastest, most efficient physical path across the internet backbone to send your request to the target server (like Google, Netflix, or a banking website).

- **The data sending process:**
 - **Your Device:** You click a link, generating digital data.
 - **The Router:** Your device sends that data to the router, which organizes your local traffic and passes the data to the modem.
 - **The Modem:** The modem takes the digital data, modulates it into a signal suitable for the line, and pushes it out of your house.
 - **The ISP:** The ISP receives the signal, demodulates it back into digital data, and routes it across the global internet infrastructure.
 - **The Destination:** The data arrives at the destination server (the website or app you are trying to reach).

If you have a router with no modem, your local devices can still talk to each other, but you will not have an internet connection. Because a router lacks the ability to translate signals for an Internet Service Provider (ISP), it cannot connect your home to the outside world on its own.

A router's primary job is actually to manage that local network (known as a Local Area Network or LAN).

The router keeps track of every device in your house by giving each one a unique local IP address (like 192.168.1.5). Because it has this map of your home, it can route data directly between your phone, your laptop, and your smart TV without ever needing to involve a modem or the internet.

So, the modem is just one of the things a router can route to.

- **Network switch:** A network switch is a piece of hardware that connects multiple devices—like computers, printers, and servers—together inside a single local area network (LAN)

Think of a switch as a smart traffic controller for data. When one device sends data to another, the switch receives the data packet and sends it only to the specific destination device, rather than broadcasting it to everything on the network

Every networked device has a unique physical hardware code called a **MAC (Media / medium Access Control) address**. It uses 12 hexadecimal characters (numbers 0–9 and letters A–F), usually split into six pairs

When data arrives at a port, the switch reads the target MAC address and consults its internal lookup table to see which port connects to that specific device

- **Router vs switch:** The device you call a "router" sitting in your living room is actually an all-in-one combo unit. It houses three entirely separate networking components stuffed into a single plastic box:
 1. **A Router:** To connect your home network to the outside internet.
 2. **A Network Switch:** The row of 4 Ethernet ports on the back, which handles the local traffic between your wired devices.
 3. **A Wireless Access Point:** The antennas that broadcast the Wi-Fi signal to your wireless devices.

The modem is what actually brings the internet into your home. It takes the raw, infrastructure signal from your ISP (whether that is traveling over fiber optic lines, coaxial cable, or phone lines) and translates it into a standard digital network signal.

A standalone modem only speaks to one single device at a time and hands out exactly one public IP address. If you plugged a laptop directly into a raw modem, that laptop would get internet, but you wouldn't be able to plug in a second device.

The router plugs directly into the modem to take that single internet connection and split it safely among all the devices in your house.

The router creates a private internal network and assigns a unique local IP address (like 192.168.1.X) to your phone, TV, computer, and smart devices.

If you decompose a network into its individual physical boxes, the sequence is

ISP → Modem → Router → Switch → Devices

- **Wireless access point (WAP):** A WiFi access point (AP) is a networking hardware device that creates a wireless local area network, usually inside a large building or home, by bridging a wired Ethernet connection to wireless signals

An AP plugs into a router or network switch using an Ethernet cable. It takes the data from the wired network and converts it into radio waves, broadcasting a WiFi signal. Phones, laptops, and smart home devices connect to this wireless signal just like they would connect to a standard router

Routers act as the "brain" of a home network. They assign IP addresses, route internet traffic, and include a built-in firewall for security

Access points focus purely on distributing a wireless signal. They do not manage traffic routing or provide firewall protection on their own; they rely on a connected router to handle those duties

- **Wifi extenders:** Plugs into an electrical outlet and repeats an existing wireless signal through the air, which can cut your available internet speed in half
- **WAN and LAN:** A LAN (Local Area Network) is a small computer network that connects devices within a single building or room, while a WAN (Wide Area Network) is a large network that connects multiple smaller networks across huge distances like cities, countries, or the entire globe
- **Pure router:** A pure, enterprise-grade router typically only features two main ports: an Internet (WAN) port to connect to the modem, and a Local (LAN) port to connect to your network.

You would run an Ethernet cable from the router's single LAN port directly into one device (like a single desktop PC). If you wanted to connect a second computer, a smart TV, or a game console, you would be entirely out of physical plugs.

Because a pure router only routes traffic between different networks (your home vs. the internet), it doesn't naturally know how to pass data sideways between devices. Even if you found a rare pure router with two LAN ports, if Computer A wanted to send a file to Computer B, the router would have to waste processing power treating them as completely separate networks rather than letting them talk instantly through a shared local switchboard.

A pure router has no wireless antennas. Without a built-in switch or a wireless access point, you wouldn't have Wi-Fi for your phone, tablet, or smart home gadgets.

You could absolutely ditch the router if you only ever need to connect one single device to the internet. If you plug a desktop computer directly into a standalone modem, it will get internet access perfectly fine.

However, you still need a router to build a local network, even if that router only has one physical plug. The router does crucial network-building jobs that a modem cannot do.

- **DHCP:** DHCP (Dynamic Host Configuration Protocol) is a network management protocol that automatically assigns an IP address and other network settings to devices on a network

DHCP uses a four-step process to connect a device, often remembered by the acronym DORA:

- **Discover:** The client device broadcasts a message on the network to find an available DHCP server.
- **Offer:** The DHCP server responds with an available IP address and network settings.
- **Request:** The client replies by formally requesting the offered IP address.
- **Acknowledge:** The server sends a final confirmation, granting the IP address for a set amount of time called a lease

Your internet provider only gives your home one single public IP address (the internet version of a mailing address).

- The Modem simply passes that single public IP address to whatever is plugged into it.
 - The Router takes that single public IP address, hides it from the world, and generates hundreds of private local IP addresses (like 192.168.1.X) for your home.
- **Dial-up:** Dial-up is an older method of connecting to the internet by making a telephone call over a standard landline

Your computer uses a device called a modem (short for modulator-demodulator) to dial the phone number of an Internet Service Provider (ISP)

When the connection starts, the modems make beeps, static, and buzzing noises as they trade signals to agree on a common communication speed

- **Guided and unguided media:** With guided media, the waves are guided along a solid medium, such as a fiber-optic cable, a twisted-pair copper wire, or a coaxial cable. With unguided media, the waves propagate in the atmosphere and in outer space, such as in a wireless LAN or a digital satellite channel.
- **Transmission rate:** To send a message from a source end system to a destination end system, the source breaks long messages into smaller chunks of data known as packets. Between source and destination, each packet travels through communication links and packet switches (for which there are two predominant types, routers and link-layer switches). Packets are transmitted over each communication link at a rate equal to the full transmission rate of the link

So, if a source end system or a packet switch is sending a packet of L bits over a link with transmission rate R bits/sec, then the time to transmit the packet is L/R seconds.

- **Store and forward transmissions:** Most packet switches use store-and-forward transmission at the inputs to the links. Store-and-forward transmission means that the packet switch must receive the entire packet before it can begin to transmit the first bit of the packet onto the outbound link
- **Output buffer:** Each packet switch has multiple links attached to it. For each attached link, the packet switch has an output buffer (also called an output queue), which stores packets that the router is about to send into that link. The output buffers play a key role in packet switching. If an arriving packet needs to be transmitted onto a link but finds the link busy with the transmission of another packet, the arriving packet must wait in the output buffer. Thus, in addition to the store-and-forward delays, packets suffer output buffer queuing delays

Since the amount of buffer space is finite, an arriving packet may find that the buffer is completely full with other packets waiting for transmission. In this case, packet loss will occur—either the arriving packet or one of the already-queued packets will be dropped.

- **Forwarding Tables and Routing Protocols:** Earlier, we said that a router takes a packet arriving on one of its attached communication links and forwards that packet onto another one of its attached communication links. But how does the router determine which link it should forward the packet onto? Packet forwarding is actually done in different ways in different types of computer networks. Here, we briefly describe how it is done in the Internet.

In the Internet, every end system has an address called an IP address. When a source end system wants to send a packet to a destination end system, the source includes the destination's IP address in the packet's header.

When a packet arrives at a router in the network, the router examines a portion of the packet's destination address and forwards the packet to an adjacent router. More specifically, each router has a forwarding table that maps destination addresses (or portions of the destination addresses) to that router's outbound links. When a packet arrives at a router, the router examines the address and searches its forwarding table, using this destination address, to find the appropriate outbound link. The router then directs the packet to this outbound link.

We'll note for now that the Internet has a number of special routing protocols that are used to automatically set the forwarding tables. A routing protocol may, for example, determine the shortest path from each router to each destination and use the shortest path results to configure the forwarding tables in the routers.

- **Circuit switching:** There are two fundamental approaches to moving data through a network of links and switches: circuit switching and packet switching. Having covered packet-switched networks in the previous subsection, we now turn our attention to circuit-switched networks.

In circuit-switched networks, the resources needed along a path (buffers, link transmission rate) to provide for communication between the end systems are reserved for the duration of the communication session between the end systems. In packet-switched networks, these resources are not reserved; a session's messages use the resources on demand and, as a consequence, may have to wait (that is, queue) for access to a communication link.

This connection is called a circuit. When the network establishes the circuit, it also reserves a constant transmission rate in the network's links (representing a fraction of each link's transmission capacity) for the duration of the connection. Since a given transmission rate has been reserved for this sender-to-receiver connection, the sender can transfer the data to the receiver at the guaranteed constant rate.

When two hosts want to communicate, the network establishes a dedicated end-to-end connection between the two hosts. Thus, in order for Host A to communicate with Host B, the network must first reserve one circuit on each of two links. In this example, the dedicated end-to-end connection uses the second circuit in the first link and the fourth circuit in the second link. Because each link has four circuits, for each link used by the end-to-end connection, the connection gets one fourth of the link's total transmission capacity for the duration of the connection.

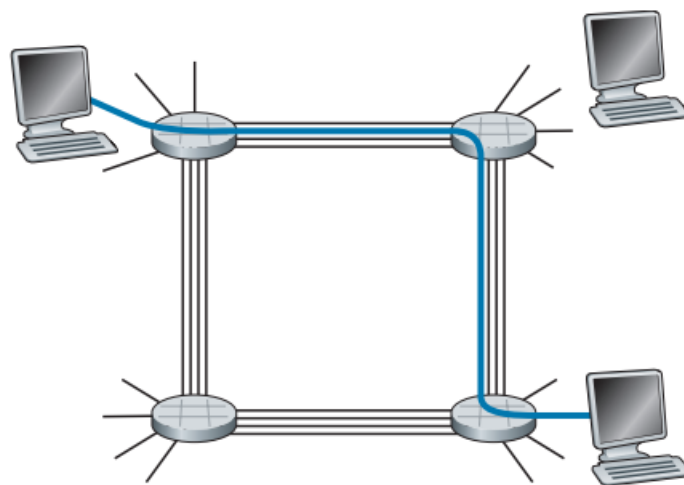


Figure 1.13 ♦ A simple circuit-switched network consisting of four switches and four links

In contrast, consider what happens when one host wants to send a packet to another host over a packet-switched network, such as the Internet. As with circuit switching, the packet is transmitted over a series of communication links. But different from circuit switching, the packet is sent into the network without reserving any link resources whatsoever. If one of the links is congested because other packets need to be transmitted over the link at the same time, then the packet will have to wait in a buffer at the sending side of the transmission link and suffer a delay. The Internet makes its best effort to deliver packets in a timely manner, but it does not make any guarantees.

- **Multiplexing in Circuit-Switched Networks:** A circuit in a link is implemented with either **frequency-division multiplexing (FDM)** or **time-division multiplexing (TDM)**. With FDM, the frequency spectrum of a link is divided up among the connections established across the link. Specifically, the link dedicates a frequency band to each connection for the duration of the connection. In telephone networks, this frequency band typically has a width of 4 kHz (that is, 4,000 hertz or 4,000 cycles per second). The width of the band is called, not surprisingly, the **bandwidth**.

For a TDM link, time is divided into frames of fixed duration, and each frame is divided into a fixed number of time slots. When the network establishes a connection across a link, the network dedicates one time slot in every frame to this connection. These slots are dedicated for the sole use of that connection, with one time slot available for use (in every frame) to transmit the connection's data.

- **Silent periods:** Proponents of packet switching have always argued that circuit switching is wasteful because the dedicated circuits are idle during silent periods. For example, when one person in a telephone call stops talking, the idle network resources (frequency bands or time slots in the links along the connection's route) cannot be used by other ongoing connections. As another example of how these resources can be underutilized, consider a radiologist who uses a circuit-switched network to remotely access a series of x-rays. The radiologist sets up a connection, requests an image, contemplates the image, and then requests a new image. Network resources are allocated to the connection but are not used (i.e., are wasted) during the radiologist's contemplation periods. Proponents of packet switching also enjoy pointing out that establishing end-to-end circuits and reserving end-to-end transmission capacity is complicated and requires complex signaling software to coordinate the operation of the switches along the end-to-end path.
- **Packet vs circuit switching:** Proponents of packet switching argue that (1) it offers better sharing of transmission capacity than circuit switching and (2) it is simpler, more efficient, and less costly to implement than circuit switching.
- **A Network of Networks:** We remark that the ISPs themselves must be interconnected. This is done by creating a network of networks—understanding this phrase is the key to understanding the Internet.

Our first network structure, Network Structure 1, interconnects all of the access ISPs with a single global transit ISP. Our (imaginary) global transit ISP is a network of routers and communication links that not only spans the globe, but also has at least one router near each of the hundreds of thousands of access ISPs. Of course, it would be very costly for the global ISP to build such an extensive network. To be profitable, it would naturally charge each of the access ISPs for connectivity, with the pricing reflecting (but not necessarily directly proportional to) the amount of traffic an access ISP exchanges with the global ISP. Since the access ISP pays the global transit ISP, the access ISP is said to be a customer and the global transit ISP is said to be a provider.

Now if some company builds and operates a global transit ISP that is profitable, then it is natural for other companies to build their own global transit ISPs and compete with the original global transit ISP. This leads to Network Structure 2, which consists of the hundreds of thousands of access ISPs and multiple global transit ISPs. The access ISPs certainly prefer Network Structure 2 over Network Structure 1 since they can now choose among the competing global transit providers as a function of their pricing and services. Note, however, that the global transit ISPs themselves must interconnect: Otherwise access ISPs connected to one of the global transit providers would not be able to communicate with access ISPs connected to the other global transit providers.

Network Structure 2, just described, is a two-tier hierarchy with global transit providers residing at the top tier and access ISPs at the bottom tier. This assumes that global transit ISPs are not only capable of getting close to each and every access ISP, but also find it economically desirable to do so. In reality, although some ISPs do have impressive global coverage and do directly connect with many access ISPs, no ISP has presence in each and every city in the world. Instead, in any given region, there may be a regional ISP to which the access ISPs in the region connect. Each regional ISP then connects to tier-1 ISPs. Tier-1 ISPs are similar to our (imaginary) global transit ISP; but tier-1 ISPs, which actually do exist, do not have a presence in every city in the world.

- **Tier-1 ISPs (The Global Backbone):** A Tier 1 ISP sits at the very top of the internet hierarchy. They own and operate the core infrastructure of the internet, including massive transoceanic fiber-optic submarine cables and terrestrial networks

They reach the entire global internet without ever paying anyone for data traffic (IP transit). Instead, they use settlement-free peering—trading data traffic freely with other Tier-1 networks on equal terms.

Examples are AT&T, Verizon, Tata Communications, and Lumen Technologies

- **Global & Regional ISPs (Tier-2 ISPs):** Tier-2 ISPs operate large regional or national networks. They have broad geographic coverage, but they cannot reach every single corner of the globe on their own

They peer for free with some networks, but they must buy IP transit from Tier-1 ISPs to pass data across the rest of the world. They often sell connectivity down to local providers or large enterprise clients

- **Access ISPs (Tier-3 ISPs):** An access ISP is the local provider that delivers the internet directly to end-users—homes, apartments, and small businesses. This is the company you pay your monthly internet bill to

They do not own a global or regional backbone. They rely entirely on Tier-2 or Tier-1 transit providers to hand off and carry local traffic out to the wider world.

- **Point of presence:** In networking, PoP stands for Point of Presence, which is a physical location or access point that connects different networks or lets users hook up to an Internet Service Provider (ISP).

It links a user's local network or device to a larger service provider's network. It usually contains routers, high-speed switches, servers, and fiber-optic multiplexers inside a data center or telecom closet

- **Multi-home:** Any ISP (except for tier-1 ISPs) may choose to multi-home, that is, to connect to two or more provider ISPs. So, for example, an access ISP may multi-home with two regional ISPs, or it may multihome with two regional ISPs and also with a tier-1 ISP. Similarly, a regional ISP may multi-home with multiple tier-1 ISPs. When an ISP multi-homes, it can continue to send and receive packets into the Internet even if one of its providers has a failure.
- **Peer:** To reduce these costs, a pair of nearby ISPs at the same level of the hierarchy can peer, that is, they can directly connect their networks together so that all the traffic between them passes over the direct connection rather than through upstream intermediaries. When two ISPs peer, it is typically settlement-free, that is, neither ISP pays the other.
- **Internet exchange point:** Along these same lines, a third-party company can create an Internet Exchange Point (IXP), which is a meeting point where multiple ISPs can peer together. An IXP is typically in a stand-alone building with its own switches
- **Content-provider network (modern):** A content-provider network, more commonly known as a Content Delivery Network (CDN) or Content Distribution Network, is a globally distributed group of servers that work together to provide fast, reliable, and secure delivery of internet content.

The primary goal of a CDN is to bridge the geographical distance between an end user and the website's original hosting server (the origin server), drastically reducing latency (the delay before a data transfer begins).

Without a CDN, every single user request must travel all the way to the origin server—even if that server is located on a different continent. This causes slow load times and risks overwhelming the website

A CDN changes this flow through caching and intelligent routing:

- **Caching Content:** The CDN replicates and saves temporary copies of the website's static content (like images, videos, HTML pages, and JavaScript files) across its entire server network
- **User Request Routing:** When a user visits a webpage or streams a video, their network request is intercepted by the CDN's smart routing algorithms
- **Edge Delivery:** Instead of sending the request to the central origin server, the CDN directs the user to the nearest physical data center—known as an Edge Server or Point of Presence (PoP). This nearby server delivers the content instantly
- **Delay, Loss, and Throughput in Packet-Switched Networks:** Recall that a packet starts in a host (the source), passes through a series of routers, and ends its journey in another host (the destination). As a packet travels from one node (host or router) to the subsequent node (host or router) along this path, the packet suffers from several types of delays at each node along the path. The most important of these delays are the **nodal processing delay**, **queuing delay**, **transmission delay**, and **propagation delay**; together, these delays accumulate to give a total nodal delay.

As part of its end-to-end route between source and destination, a packet is sent from the upstream node through router A to router B. Our goal is to characterize the nodal delay at router A. Note that router A has an outbound link leading to router B. This link is preceded by a queue (also known as a buffer). When the packet arrives at router A from the upstream node, router A examines the packet's header to determine the appropriate outbound link for the packet and then directs the packet to this link. In this example, the outbound link for the packet is the one that leads to router B. A packet can be transmitted on a link only if there is no other packet currently being transmitted on the link and if there are no other packets preceding it in the queue; if the link is currently busy or if there are other packets already queued for the link, the newly arriving packet will then join the queue.

- **Processing Delay:** The time required to examine the packet's header and determine where to direct the packet is part of the processing delay. The processing delay can also include other factors, such as the time needed to check for bit-level errors in the packet that occurred in transmitting the packet's bits from the upstream node to router A. Processing delays in high-speed routers are typically on the order of microseconds or less. After this nodal processing, the router directs the packet to the queue that precedes the link to router B.
- **Queuing Delay:** At the queue, the packet experiences a queuing delay as it waits to be transmitted onto the link. The length of the queuing delay of a specific packet will depend on the number of earlier-arriving packets that are queued and waiting for transmission onto the link.
- **Transmission Delay:** Assuming that packets are transmitted in a first-come-first-served manner, as is common in packet-switched networks, our packet can be transmitted only after all the packets that have arrived before it have been transmitted.

The **transmission delay** is L/R , where L is the length in bits, and R is the number of bits transmitted per second. This is the amount of time required to push (that is, transmit) all of the packet's bits into the link. Transmission delays are typically on the order of microseconds to milliseconds in practice.

- **Propagation Delay:** Once a bit is pushed into the link, it needs to propagate to router B. The time required to propagate from the beginning of the link to router B is the propagation delay. The bit propagates at the propagation speed of the link. The propagation speed depends on the physical medium of the link

The propagation delay is the distance between two routers divided by the propagation speed. That is, the propagation delay is d/s , where d is the distance between router A and router B and s is the propagation speed of the link. Once the last bit of the packet propagates to node B, it and all the preceding bits of the packet are stored in router B. The whole process then continues with router B now performing the forwarding. In wide-area networks, propagation delays are on the order of milliseconds.

- **Traffic intensity:** When is the queuing delay large and when is it insignificant? The answer to this question depends on the rate at which traffic arrives at the queue, the transmission rate of the link, and the nature of the arriving traffic, that is, whether the traffic arrives periodically or arrives in bursts. To gain some insight here, let a denote the average rate at which packets arrive at the queue (a is in units of packets/sec). Recall that R is the transmission rate; that is, it is the rate (in bits/sec) at which bits are pushed out of the queue. Also suppose, for simplicity, that all packets consist of L bits. Then the average rate at which bits arrive at the queue is La bits/sec. Finally, assume that the queue is very big, so that it can hold essentially an infinite number of bits. The ratio La/R is called the **traffic intensity**.

If $\lambda a/R > 1$, then the average rate at which bits arrive at the queue exceeds the rate at which the bits can be transmitted from the queue. In this unfortunate situation, the queue will tend to increase without bound and the queuing delay will approach infinity! Therefore, one of the golden rules in traffic engineering is: Design your system so that the traffic intensity is no greater than 1.

- **Packet Loss:** In our discussions above, we have assumed that the queue is capable of holding an infinite number of packets. In reality a queue preceding a link has finite capacity, although the queuing capacity greatly depends on the router design and cost. Because the queue capacity is finite, packet delays do not really approach infinity as the traffic intensity approaches 1. Instead, a packet can arrive to find a full queue. With no place to store such a packet, a router will drop that packet; that is, the packet will be lost.

From an end-system viewpoint, a packet loss will look like a packet having been transmitted into the network core but never emerging from the network at the destination. The fraction of lost packets increases as the traffic intensity increases. Therefore, performance at a node is often measured not only in terms of delay, but also in terms of the probability of packet loss

- **Traceroute:** To get a hands-on feel for end-to-end delay in a computer network, we can make use of the Traceroute program. Traceroute is a simple program that can run in any Internet host. When the user specifies a destination hostname, the program in the source host sends multiple, special packets toward that destination. As these packets work their way toward the destination, they pass through a series of routers. When a router receives one of these special packets, it sends back to the source a short message that contains the name and address of the router.

More specifically, suppose there are $N - 1$ routers between the source and the destination. Then the source will send N special packets into the network, with each packet addressed to the ultimate destination. These N special packets are marked 1 through N , with the first packet marked 1 and the last packet marked N . When the n th router receives the n th packet marked n , the router does not forward the packet toward its destination, but instead sends a message back to the source. When the destination host receives the N th packet, it too returns a message back to the source. The source records the time that elapses between when it sends a packet and when it receives the corresponding return message; it also records the name and address of the router (or the destination host) that returns the message. In this manner, the source can reconstruct the route taken by packets flowing from source to destination, and the source can determine the round-trip delays to all the intervening routers

- **Throughput in Computer Networks:** In addition to delay and packet loss, another critical performance measure in computer networks is end-to-end throughput. To define throughput, consider transferring a large file from Host A to Host B across a computer network. This transfer might be, for example, a large video clip from one computer to another. The instantaneous throughput at any instant of time is the rate (in bits/sec) at which Host B is receiving the file

If the file consists of F bits and the transfer takes T seconds for Host B to receive all F bits, then the **average throughput** of the file transfer is F/T bits/sec.